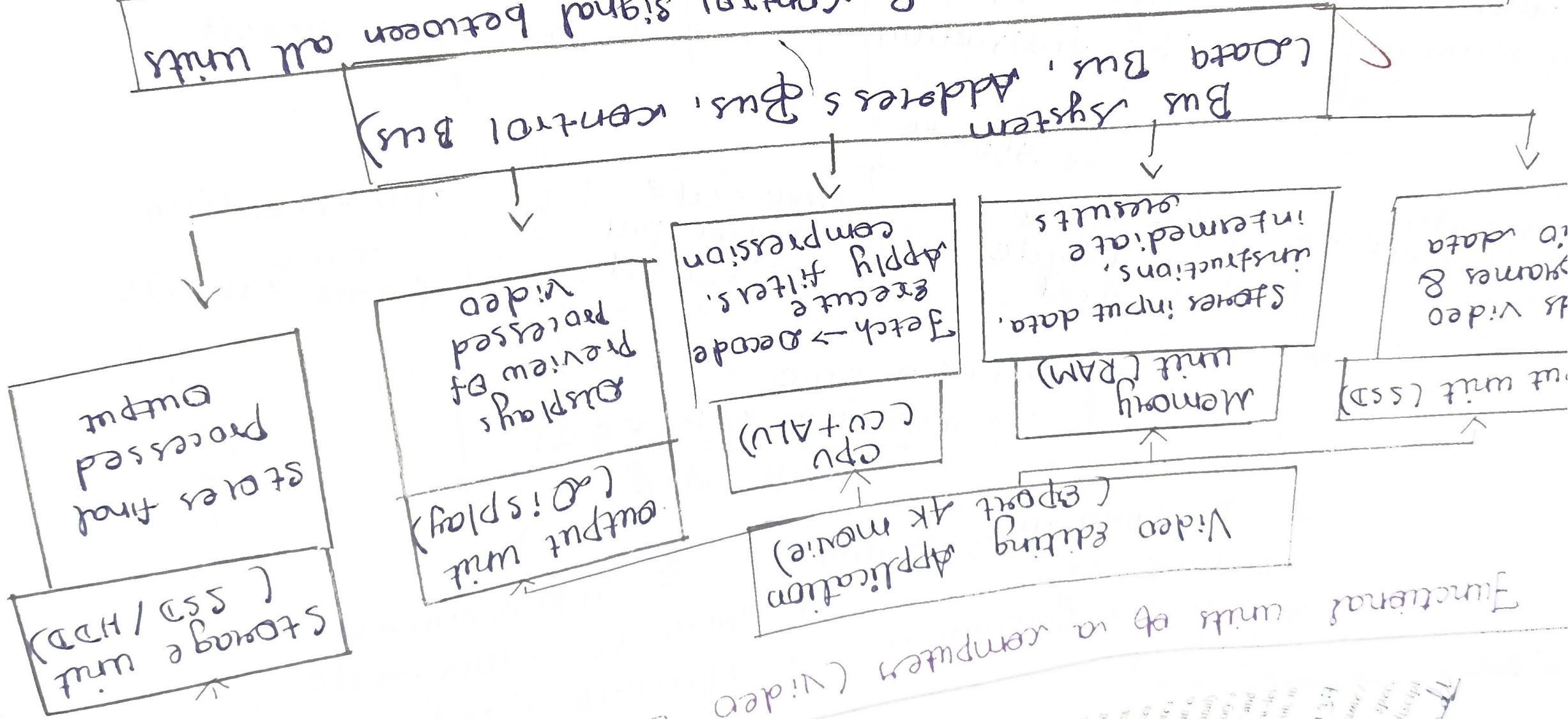


Editing Application

Functional units of a computer (Video Editing Application)



Transfer Data, Instruction & control signal between all units

coordination of 5 functional units

Produce

Retrieve Data -> Process Video & audio -> final output

A video
an NMI
a concept
How do the
the final output
A software engi
identify the cause
and examines perf
from linking the instr
influence CPU execu
performance
A robotics engineer is sel
architectures is sel
comparing to determine
organization, and memory
action's access, multithreading
software development
different operations is optimiz
CPU Register, indirect, and indir
fact, addressing mode influences
side real world applications where
how are engineer is debugging an
instruction representation where
is converted between binary and
that notation is preferred in binary
support instruction representation in
binary

15/12/2021 10:07:17 PM

Name: _____

Guidelines spec
rules will result in discipli

Student: _____

Instruction Execution cycle & Performance parameters

Instruction execution cycle

Fetch stage
Fetch instruction
from memory

Decode stage
Decodes & interprets
the instruction

Execute stage
Execute operation
& store result

Affects

CPI
Clock cycle
per instruction

Clock cycles
Total cycle
used

Clock rate
Cycle per
second

CPU Execution
Time
(Time taken)

$$\text{CPU Execution Time} = \frac{\text{Instruction} \times \text{CPI} \times \text{CCT}}{\text{clock rate}}$$

Performance Evaluation & Improvement

- * Reduce CPI
- * Optimize Instruction

- * Increase clock rate
- * Efficient Hardware

Comparison of 8085 and 8086

Microprocessor comparison

3.

8085	Feature	8086
8 bit Microprocessor	Architecture	16-bit Microprocessor
74 Instructions	Instruction set	- 130 Instructions
6 General purposes (B, C, D, E, H, L) + 2 (PC, SP)	Registers	8 General purpose + 4 segment
16-bit (64 KB Memory)	Address Bus	20-bit (1 MB addressable Memory)
Simple Memory	Memory organization	segment Memory
5 Addressing Modes	Addressing Modes	12 Addressing Modes
Slower	Speed & performance	Faster
Simple	Application	Multitasking & complex Real-time Application

Addressing Modes and their Applications

Addressing Modes

Immediate Address
Data is part of instruction

Direct Address
Address is given in instruction

Register Address
Operand in register

Indirect Address
is register

Indexed Address
 $\text{Address} + \text{index value}$

used in instruction

Data transfer

Arithmetic

control

Influence on performance

Memory Access time

Instruction size

Execution speed

processor efficiency

Real-world Application

Immediate

Direct

Register

Indirect

Indexed

5. Binary and Hexadecimal Number system in computer system

